

3.1 Factors and Multiples of Whole Numbers

LESSON FOCUS

Determine prime factors, greatest common factors, and least common multiples of whole numbers.

The origins of these belts are unknown. They are on display in the MacBride museum in Whitehorse, Yukon.



Make Connections

In the belts above, the patterns are 12 beads long and 40 beads long. How many beads long must a belt be for it to be created using either pattern?

Construct Understanding

TRY THIS

Work with a partner.

- A.** List some powers of 2. Make another list of powers of 3.
Pick a number from each list and multiply them to create a different number. What are the factors of this number?
What are some multiples of this number?
- B.** Compare your number to your partner's number.
Which factors do the two numbers have in common?
Which factor is the greatest?
- C.** What are some multiples the two numbers have in common?
Which multiple is the least?
- D.** How can you use the product of powers from Step A to determine the greatest factor and the least multiple that the numbers have in common?

When a factor of a number has exactly two divisors, 1 and itself, the factor is a *prime factor*. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12. The prime factors of 12 are 2 and 3. To determine the **prime factorization** of 12, write 12 as a product of its prime factors: $2 \times 2 \times 3$, or $2^2 \times 3$

To avoid confusing the multiplication symbol with the variable x , we use a dot to represent the multiplication operation: $12 = 2 \cdot 2 \cdot 3$, or $2^2 \cdot 3$

The first 10 prime numbers are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
Natural numbers greater than 1 that are not prime are *composite*.

The **prime factorization** of a natural number is the number written as a product of its prime factors.

Every composite number can be expressed as a product of prime factors.

Example 1 Determining the Prime Factors of a Whole Number

Write the prime factorization of 3300.

SOLUTIONS

Method 1

Draw a factor tree.

Write 3300 as a product of 2 factors.

Both 33 and 100 are composite numbers, so we can factor again.

Both 3 and 11 are prime factors, but 4 and 25 can be factored further.

The prime factors of 3300 are 2, 3, 5, and 11.

The prime factorization of 3300 is: $2 \cdot 2 \cdot 3 \cdot 5 \cdot 5 \cdot 11$, or $2^2 \cdot 3 \cdot 5^2 \cdot 11$

Method 2

Use repeated division by prime factors.

Begin by dividing 3300 by the least prime factor, which is 2.

Divide by this prime factor until it is no longer a factor.

Continue to divide each quotient by a prime factor until the quotient is 1.

$$3300 \div 2 = 1650$$

$$1650 \div 2 = 825$$

$$825 \div 3 = 275$$

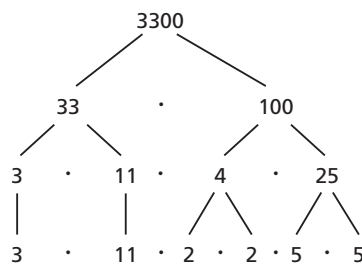
$$275 \div 5 = 55$$

$$55 \div 5 = 11$$

$$11 \div 11 = 1$$

The prime factors of 3300 are 2, 3, 5, and 11.

The prime factorization of 3300 is: $2 \cdot 2 \cdot 3 \cdot 5 \cdot 5 \cdot 11$, or $2^2 \cdot 3 \cdot 5^2 \cdot 11$



CHECK YOUR UNDERSTANDING

- Write the prime factorization of 2646.

[Answer: $2 \cdot 3^3 \cdot 7^2$]

What other factor trees could you draw?

Could we have divided by the prime factors in a different order? If so, which order do you think is best? Why?

The **greatest common factor** of two or more numbers is the greatest factor the numbers have in common.

For 2 or more natural numbers, we can determine their **greatest common factor**.

Example 2 Determining the Greatest Common Factor

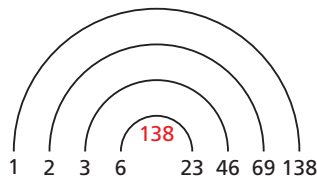
Determine the greatest common factor of 138 and 198.

SOLUTIONS

Method 1

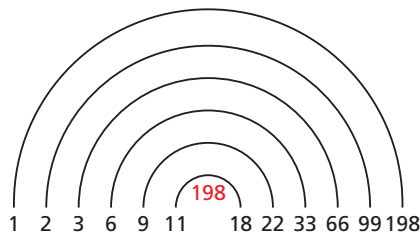
Use division facts to determine all the factors of each number.
Record the factors as a “rainbow.”

$$\begin{aligned} 138 \div 1 &= 138 \\ 138 \div 2 &= 69 \\ 138 \div 3 &= 46 \\ 138 \div 6 &= 23 \end{aligned}$$



Since 23 is a prime number, there are no more factors of 138.

$$\begin{aligned} 198 \div 1 &= 198 \\ 198 \div 2 &= 99 \\ 198 \div 3 &= 66 \\ 198 \div 6 &= 33 \\ 198 \div 9 &= 22 \\ 198 \div 11 &= 18 \end{aligned}$$



There are no more factors of 198 between 11 and 18.

The common factors of 138 and 198 are: 1, 2, 3, and 6.

So, the greatest common factor is 6.

Method 2

Check to see which factors of 138 are also factors of 198.

Start with the greatest factor.

The factors of 138 are: 1, 2, 3, 6, 23, 46, 69, 138

198 is not divisible by 138, 69, 46, or 23.

198 is divisible by 6: $198 \div 6 = 33$

The greatest common factor is 6.

Method 3

Write the prime factorization of each number.

Highlight the factors that appear in each prime factorization.

$$138 = 2 \cdot 3 \cdot 23$$

$$198 = 2 \cdot 3 \cdot 3 \cdot 11$$

The greatest common factor is $2 \cdot 3$, which is 6.

CHECK YOUR UNDERSTANDING

- Determine the greatest common factor of 126 and 144.

[Answer: 18]

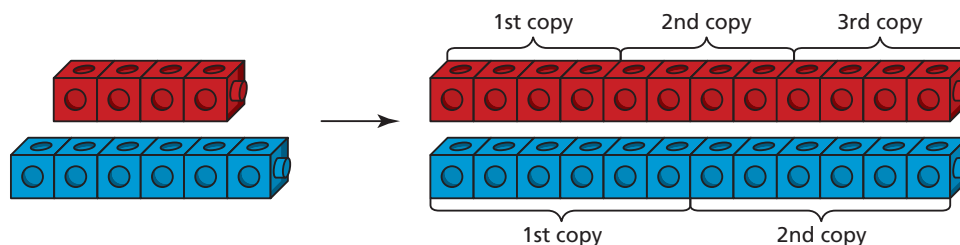
How would you determine the greatest common factor of 3 numbers?

To generate multiples of a number, multiply the number by the natural numbers; that is, 1, 2, 3, 4, 5, and so on. For example, some multiples of 26 are:

$$26 \cdot 1 = 26 \quad 26 \cdot 2 = 52 \quad 26 \cdot 3 = 78 \quad 26 \cdot 4 = 104$$

For 2 or more natural numbers, we can determine their **least common multiple**.

We can determine the least common multiple of 4 and 6 by combining identical copies of each smaller chain to create two chains of equal length.



The shortest chain possible is 12 cubes long.
So, the least common multiple of 4 and 6 is 12.

The **least common multiple** of two or more numbers is the least number that is divisible by each number.

Example 3 Determining the Least Common Multiple

Determine the least common multiple of 18, 20, and 30.

SOLUTIONS

Method 1

List the multiples of each number until the same multiple appears in all 3 lists.

Multiples of 18 are: 18, 36, 54, 72, 90, 108, 126, 144, 162, **180**, ...

Multiples of 20 are: 20, 40, 60, 80, 100, 120, 140, 160, **180**, ...

Multiples of 30 are: 30, 60, 90, 120, 150, **180**, ...

The least common multiple of 18, 20, and 30 is 180.

Method 2

Check to see which multiples of 30 are also multiples of 18 and 20.

The multiples of 30 are: 30, 60, 90, 120, 150, 180, ...

18 divides exactly into: 90, **180**, ...

20 divides exactly into: 60, 120, **180**, ...

The least common multiple of 18, 20, and 30 is 180.

(Solution continues.)

CHECK YOUR UNDERSTANDING

- Determine the least common multiple of 28, 42, and 63.

[Answer: 252]

Method 3

Write the prime factorization of each number.

Highlight the greatest power of each prime factor in any list.

$$18 = 2 \cdot 3 \cdot 3 = 2 \cdot 3^2$$

The greatest power of 3 in any list is 3^2 .

$$20 = 2 \cdot 2 \cdot 5 = 2^2 \cdot 5$$

The greatest power of 2 in any list is 2^2 .

$$30 = 2 \cdot 3 \cdot 5$$

The greatest power of 5 in any list is 5.

The least common multiple is the product of the greatest power of each prime factor:

$$\begin{aligned} 2^2 \cdot 3^2 \cdot 5 &= 4 \cdot 9 \cdot 5 \\ &= 180 \end{aligned}$$

The least common multiple of 18, 20, and 30 is 180.

Are there any pairs of numbers for which the least common multiple and greatest common factor are the same number? Explain.

Example 4

Solving Problems that Involve Greatest Common Factor and Least Common Multiple

- a) What is the side length of the smallest square that could be tiled with rectangles that measure 16 cm by 40 cm? Assume the rectangles cannot be cut. Sketch the square and rectangles.
- b) What is the side length of the largest square that could be used to tile a rectangle that measures 16 cm by 40 cm? Assume that the squares cannot be cut. Sketch the rectangle and squares.

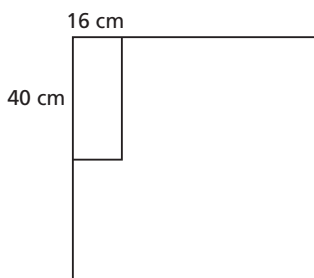
SOLUTION

- a) In the square, arrange all the rectangles with the same orientation.

The shorter side of each rectangle measures 16 cm. So, the side length of the square must be a multiple of 16.

The longer side of each rectangle measures 40 cm. So, the side length of the square must be a multiple of 40.

So, the side length of the square must be a common multiple of 16 and 40.



(Solution continues.)

CHECK YOUR UNDERSTANDING

4. a) What is the side length of the smallest square that could be tiled with rectangles that measure 8 in. by 36 in.? Assume the rectangles cannot be cut. Sketch the square and rectangles.
- b) What is the side length of the largest square that could be used to tile a rectangle that measures 8 in. by 36 in.? Assume that the squares cannot be cut. Sketch the rectangle and squares.

[Answers: a) 72 in. b) 4 in.]

The side length of the smallest square will be the least common multiple of 16 and 40.

Write the prime factorization of each number. Highlight the greatest power of each prime factor in either list.

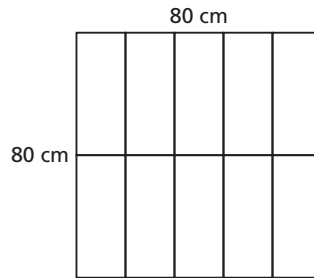
$$16 = 2^4$$

$$40 = 2^3 \cdot 5$$

The least common multiple is:

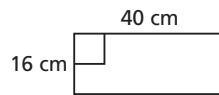
$$2^4 \cdot 5 = 80$$

The side length of the smallest square is 80 cm.



In *Example 4a*, would the answer change if the rectangles could be oriented in any direction? Explain.

- b) The shorter side of the rectangle measures 16 cm. So, the side length of the square must be a factor of 16.



The longer side of the rectangle measures 40 cm. So, the side length of the square must be a factor of 40.

So, the side length of the square must be a common factor of 16 and 40.

The side length of the largest square will be the greatest common factor of 16 and 40.

Write the prime factorization of each number.

Highlight the prime factors that appear in both lists.

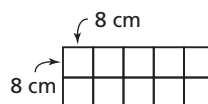
$$16 = 2 \cdot 2 \cdot 2 \cdot 2$$

$$40 = 2 \cdot 2 \cdot 2 \cdot 5$$

The greatest common factor is:

$$2 \cdot 2 \cdot 2 = 8$$

The largest square has side length 8 cm.



Why is it helpful to write the prime factorization of 16 and 40 as a product of powers in part a and as an expanded product in part b?

Discuss the Ideas

1. What strategies can you use to determine the prime factorization of a whole number?
2. How can you use the prime factorization of a number to determine all the factors of that number?

Exercises

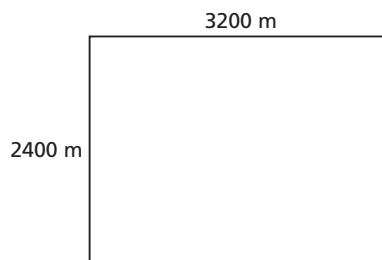
A

3. List the first 6 multiples of each number.
 a) 6 b) 13 c) 22
 d) 31 e) 45 f) 27
4. List the prime factors of each number.
 a) 40 b) 75 c) 81
 d) 120 e) 140 f) 192
5. Write each number as a product of its prime factors.
 a) 45 b) 80 c) 96
 d) 122 e) 160 f) 195

B

6. Use powers to write each number as a product of its prime factors.
 a) 600 b) 1150
 c) 1022 d) 2250
 e) 4500 f) 6125
7. Explain why the numbers 0 and 1 have no prime factors.
8. Determine the greatest common factor of each pair of numbers.
 a) 46, 84 b) 64, 120
 c) 81, 216 d) 180, 224
 e) 160, 672 f) 220, 860
9. Determine the greatest common factor of each set of numbers
 a) 150, 275, 420 b) 120, 960, 1400
 c) 126, 210, 546, 714 d) 220, 308, 484, 988
10. Determine the least common multiple of each pair of numbers.
 a) 12, 14 b) 21, 45
 c) 45, 60 d) 38, 42
 e) 32, 45 f) 28, 52
11. Determine the least common multiple of each set of numbers.
 a) 20, 36, 38 b) 15, 32, 44
 c) 12, 18, 25, 30 d) 15, 20, 24, 27
12. Explain the difference between determining the greatest common factor and the least common multiple of 12 and 14.

13. Two marching bands are to be arranged in rectangular arrays with the same number of columns. One band has 42 members, the other has 36 members. What is the greatest number of columns in the array?
14. When is the product of two numbers equal to their least common multiple?
15. How could you use the greatest common factor to simplify a fraction? Use this strategy to simplify these fractions.
 a) $\frac{185}{325}$ b) $\frac{340}{380}$ c) $\frac{650}{900}$
 d) $\frac{840}{1220}$ e) $\frac{1225}{2750}$ f) $\frac{2145}{1105}$
16. How could you use the least common multiple to add, subtract, or divide fractions? Use this strategy to evaluate these fractions.
 a) $\frac{9}{14} + \frac{11}{16}$ b) $\frac{8}{15} + \frac{11}{20}$
 c) $\frac{5}{24} - \frac{1}{22}$ d) $\frac{9}{10} + \frac{5}{14} + \frac{4}{21}$
 e) $\frac{9}{25} + \frac{7}{15} - \frac{5}{8}$ f) $\frac{3}{5} - \frac{5}{18} + \frac{7}{3}$
 g) $\frac{3}{5} \div \frac{4}{9}$ h) $\frac{11}{6} \div \frac{2}{7}$
17. A developer wants to subdivide this rectangular plot of land into congruent square pieces. What is the side length of the largest possible square?



18. Do all whole numbers have at least one prime factor? Explain.
19. a) What are the dimensions of the smallest square that could be tiled using an 18-cm by 24-cm tile? Assume the tiles cannot be cut.
 b) Could the tiles in part a be used to cover a floor with dimensions 6.48 m by 15.12 m? Explain.

20. The Dominion Land Survey is used to divide much of western Canada into sections and acres. One acre of land is a rectangle measuring 66 feet by 660 feet.

- A section is a square with side length 1 mile. Do the rectangles for 1 acre fit exactly into a section? Justify your answer.
[1 mile = 5280 feet]
- A quarter section is a square with side length $\frac{1}{2}$ mile. Do the rectangles for 1 acre fit exactly into a quarter section? Justify your answer.
- What is the side length of the smallest square into which the rectangles for 1 acre will fit exactly?

C

- Marcia says that she knows that 61 is a prime number because she tried dividing 61 by all the natural numbers up to and including 7, and none of them was a factor. Do you agree with Marcia? Explain.
- A bar of soap has the shape of a rectangular prism that measures 10 cm by 6 cm by 3 cm. What is the edge length of the smallest cube that could be filled with these soap bars?

Reflect

Describe strategies you would use to determine the least common multiple and the greatest common factor of a set of numbers.



THE WORLD OF MATH

Math Fact: Cryptography

Cryptography is the art of writing or deciphering messages in code. Cryptographers use a key to encode and decode messages. One way to generate a key is to multiply two large prime numbers. This makes it almost impossible to decipher the code without knowing the original numbers that were multiplied to encipher the message.

In 2006, mathematicians announced they had factored a 274-digit number as the product of a 120-digit prime number and a 155-digit prime number.

$$c_{274} = \frac{6^{353} - 1}{5}$$

= 9736915051844164425659589830765310381017746994454460344424676734039701450849424662984652946941
8789179481605188614420406622642320616708178468189806366368550930451357370697905234613513066631
78231611242601530501649312653193616879609578238789980474856787874287635916569919566643
= p120 x p155
= 1350952613301126518307750496355908073811210311113827323183908467597440721656365429201433517381
98057636666351316191686483 x 7207443811113019376439358640290253916138908670997078170498495662717
8573407484509481161087627373286704178679466051451768242073072242783688661390273684623521